Supplementary Material:
Information-limited fidelity estimation of composed quantum channels: when to compose, measure, or learn

This supplement reports oracle ground-truth verification (S1), extended distribution-quality metrics and reliability diagnostics (S2), per-seed sample-complexity tables for DFE and Monte Carlo (S3), ablation tables (S4), and extended length-OOD diagnostics (S5). All numbers are reproducible from the released code described in the Data and code availability statement of the main paper.

S1 Oracle ground-truth verification

For the Markovian regimes, ground-truth entanglement fidelity F_e follows from exact channel composition and the closed-form identity-target formula. Collision labels use exact joint propagation of the system and bath. Bath retention between steps is set by η. We verify three properties of the Markovian pipeline.

Associativity of Choi composition. We draw 2000 random channel triples $\Lambda_a, \Lambda_b, \Lambda_c$. Half use $d = 2$ and half use $d = 4$. We compare the two orderings $(\Lambda_a \star \Lambda_b) \star \Lambda_c$ and $\Lambda_a \star (\Lambda_b \star \Lambda_c)$. The maximum elementwise discrepancy is below 2×10^{-13}, which is at the numerical floor. The corresponding F_e values agree within 10^{-10}. The released test suite includes this check.

Cross-validation against an independent simulator. For 1,024 Markovian sequences of mixed amplitude-damping, depolarising, and phase-damping channels with lengths $L \in \{2, 4, 8, 16, 24\}$ we compare our F_e against the value obtained by reconstructing the same superoperators in QuTiP and invoking `qutip.process_fidelity`. The maximum discrepancy is 2.4×10^{-8}, the mean is 4.3×10^{-9}, and the median is 1.1×10^{-15} (i.e. most pairs agree to floating-point precision). This check provides an independent validation of the composition pipeline. The corresponding unit test uses 256 random sequences and gives identical statistics to within seed noise.

CPTP sanity at every channel. Every channel produced by the channel libraries passes a CPTP test that checks (i) Choi positive semi-definiteness via Cholesky with tolerance 10^{-9}, (ii) trace preservation $\mathrm{Tr}_\mathrm{out}\, J(\Lambda) = I$ to the same tolerance. The test runs as part of the standard validation suite and is required to succeed before any data-generation script is run.

S2 Extended calibration diagnostics

Table S1 reports distributional metrics for the collision regime. The legacy `family_ood` filename refers to the high-retention interval $\eta \in [0.85, 0.99]$. The collision family is shared with training. Two figures provide additional diagnostics. The first compares expected calibration error (ECE) before and after split-conformal recalibration on ID. The second gives reliability diagrams for ID and the $n = 48$ length-OOD endpoint.

Calibration interpretation. The ID ECE is 0.004, below the threshold of 0.05 used in our analysis. ECE rises to 0.39 and 0.50 on the two OOD splits. Conformal offsets fitted on ID transfer poorly across these shifts. FidelityNO+cal uses a separate affine correction fitted on labelled OOD data. The MAE column is measured before that correction and matches the raw FidelityNO result up to rounding. Synthetic OOD labels come from exact joint propagation of the system and bath. A hardware experiment would obtain them through measurement or characterisation. Section S3 counts the cost of finite-shot DFE labels.

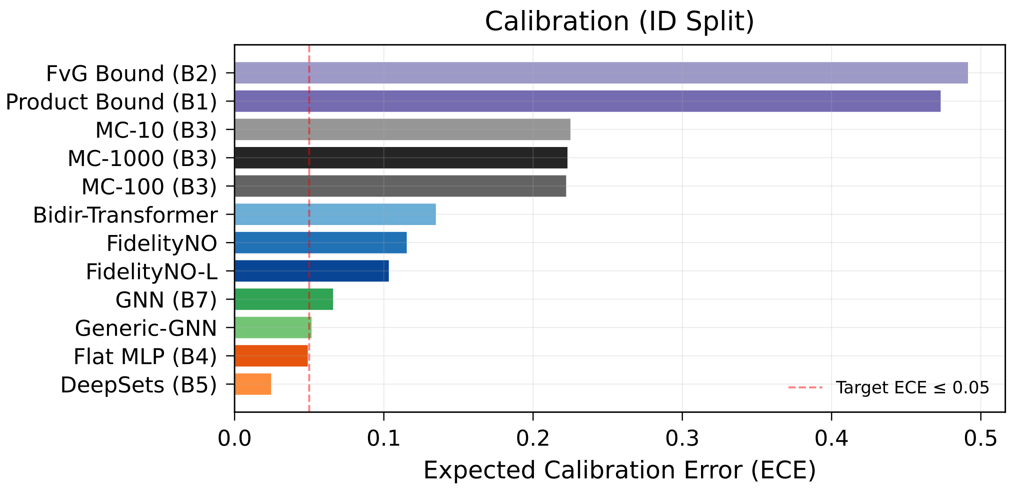

Figure S1: ECE before and after split-conformal recalibration on the in-distribution test split. The quantile-head neural surrogates improve after recalibration, with FidelityNO reaching the low-ECE region used in the main-paper comparison.

Table S1: Extended pre-affine distribution metrics for FidelityNO on the collision regime. Additive conformal offsets are fitted on `calib.npz`. MAE gives the raw point error before the labelled-OOD affine correction used in the main collision table. Results cover five seeds and 1024 sequences per split.

Split	MAE	Pinball	CRPS	ECE
`id_test`	0.025	0.009	0.018	**0.004**
`length_ood`	0.138	0.060	0.119	0.394
`family_ood`	0.383	0.179	0.359	0.500

Reliability diagram. The released figures show empirical coverage for each predicted quantile. ID curves follow the diagonal closely. On OOD data, lower quantiles tend to over-cover and upper quantiles tend to under-cover. This pattern agrees with the mean shift measured in the affine-recalibration analysis.

S3 Corrected DFE and calibration-cost audit

The first DFE implementation sampled Pauli settings with replacement and assigned 200 shots to each draw. This estimator is unbiased and has extra protocol variance for the single-qubit identity target, whose relevance support contains four Pauli settings. The revised implementation enumerates the support, allocates a fixed total budget across settings, and samples exact binomial $\{+1, -1\}$ outcomes. Table S2 reports the low-shot curve on 1024 memory-strength-OOD sequences.

Table S2: Stratified DFE on the collision memory-strength-OOD split. The estimator acts on the true composed process.

total shots/sequence	MAE in F_e	RMSE in F_e
4	0.282	0.328
8	0.183	0.227
16	0.135	0.167
32	0.092	0.113
64	0.065	0.082
128	0.044	0.055
256	0.032	0.041
512	0.023	0.029
1024	0.016	0.020

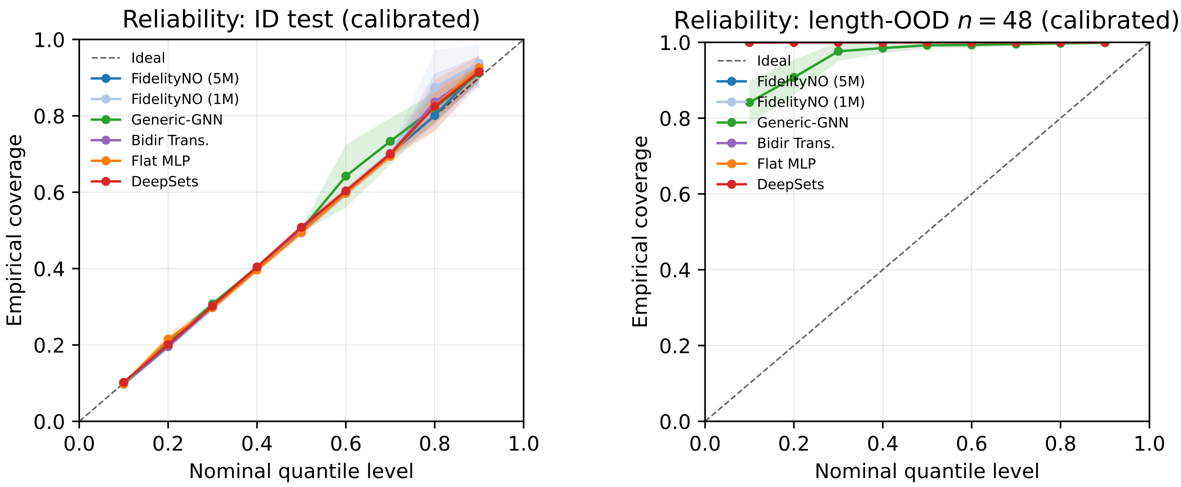

Figure S2: Reliability diagrams after split-conformal recalibration. The left panel shows ID, where the neural surrogates follow the diagonal and agree with the low ECE values in Table S1. The right panel shows length OOD at $n = 48$. All methods become conservative because conformal offsets fitted on $n \leq 16$ overstate uncertainty under this length shift.

At 2000 shots per sequence, stratified DFE gives MAE 0.0069 on ID, 0.0112 on length OOD, and 0.0116 on memory-strength OOD. These values replace the earlier with-replacement curve. The bidirectional model with $N_{\text{cal}} = 64$ reaches MAE about 0.087. Stratified DFE approaches this error near 32 shots per query and gives a lower error at 64 shots.

Finite-shot calibration labels. The main text also audits the one-time cost of labelled OOD calibration. We estimate every calibration label by stratified DFE, fit the same affine map, and evaluate against exact held-out labels. Table S3 pools five bidirectional checkpoints and five independent measurement repetitions. With 64 labels, 64 shots per label are enough for the calibration-noise penalty to be negligible.

Table S3: Bidirectional model with $N_{\text{cal}} = 64$ finite-shot calibration labels. The exact-label reference MAE is 0.087.

shots/label	total offline shots	test MAE (mean $\pm$ std)
16	1024	0.092 ± 0.007
32	2048	0.091 ± 0.006
64	4096	0.087 ± 0.003
128	8192	0.085 ± 0.001
512	32768	0.086 ± 0.001

Reproduction protocol. Run `scripts/eval_dfe.py` with `-strategy stratified -M 1` and the total-shot values above. The companion noisy-calibration evaluation script reproduces the calibration audit. Both scripts record the seed, evaluated sample count, and total shot budget in CSV output.

Marginal Monte Carlo. Classical Kraus-path sampling uses reset-bath marginals and has no observation of the retained bath. Increasing the path count reduces sampling variance around the marginal-composed process. The bias from memory strength remains.

S4 Additional ablations

We test the auxiliary physics-consistency weight, the distribution head, and the channel representation. The main paper reports the comparison between causal and bidirectional attention. All variants use the same collision data, optimiser, and curriculum. The default model uses five seeds and each ablation uses three. Every split contains 1024 sequences.

Auxiliary physics-consistency loss. The default model uses an auxiliary head predicting the purity and trace of the composed Choi, with weight $\lambda_{\text{aux}} = 0.1$. Table S4 reports collision-regime raw (uncalibrated) MAE with this loss off, default, and at heavier weight.

Table S4: Auxiliary physics-consistency loss ablation. The table reports pooled raw MAE on the collision regime and ID ECE for distributional calibration.

λ_{aux}	seeds	`id_test`	`length_ood`	`family_ood`	ECE (`id_test`)
0.0	3	0.0240	0.1390	0.3867	0.021
0.1 (default)	5	0.0247	0.1377	0.3869	0.018
0.5	3	0.0238	0.1384	0.3881	0.020

The three auxiliary-loss settings differ by at most 0.0014 across the collision splits, which lies within seed variation. We keep $\lambda_{\text{aux}} = 0.1$ for consistency with the earlier runs. Removing the auxiliary loss produces similar point-prediction MAE.

Quantile head versus Gaussian head. Table S5 compares the primary 9-quantile pinball head against a parametric truncated-Gaussian head trained with NLL.

Table S5: Distribution-head ablation. Pooled raw MAE and ID ECE on the collision regime.

Head	seeds	`id_test`	`length_ood`	`family_ood`	ECE (`id_test`)
Quantile (default)	5	0.0247	0.1377	0.3869	0.018
Gaussian	3	0.0248	0.1369	0.3876	0.039

The two heads give similar point-prediction MAE. Their calibration differs on ID. The Gaussian head has ECE 0.039, compared with 0.018 for the quantile head. We use the quantile head for its lower calibration error.

Choi versus Pauli-transfer-matrix encoder. We also test a Pauli-transfer-matrix (PTM) encoder. For a single-qubit channel, the Choi feature has 32 real and imaginary entries. The PTM has 16 real entries given by $R_{ij} = (1/d)\mathrm{Tr}[P_i \Lambda(P_j)]$. We zero-pad the PTM to the same input width. A precomputed Choi-to-PTM tensor performs the conversion in one contraction. Table S6 reports the collision MAE for both encoders.

Table S6: Channel-encoder ablation: Choi versus Pauli transfer matrix. All other architecture choices are identical.

Encoder	seeds	`id_test`	`length_ood`	`family_ood`	ECE (`id_test`)
Choi (default)	5	0.0247	0.1377	0.3869	0.018
PTM	3	0.0247	0.1356	0.3863	0.018

For $d = 2$, the two encoders differ by at most 0.0021 across all splits and metrics. This difference lies within seed variation. We use Choi throughout the paper so that the representation stays unchanged between the single-qubit and $d = 4$ experiments. At $d = 4$, the PTM contains 256 real values per channel and the real-imaginary Choi feature contains 512. The single-qubit ablation shows similar accuracy and calibration for the two choices. The released implementation documents the PTM encoder, and the reported $d = 4$ runs use Choi.

Interpretation of the ablations. The collision regime contains bath-mediated correlations across time steps. A causal transformer processes channel tokens, and the OOD affine fit corrects the deployment shift. Within this configuration, the auxiliary loss and the channel representation have little effect on MAE. The distribution head mainly affects calibration.

S5 Length-OOD diagnostics

The main paper reports pooled affine-recalibrated MAE of 0.077 for length OOD with $L \in \{24, 32, 48\}$. The following diagnostic reports each length before the pooled affine correction and shows how the raw bias changes with sequence length.

Table S7: Pre-affine length-by-length diagnostic for the collision regime. Results cover five seeds and 1024 sequences per length. The legacy sampling table used a different subset. Corrected pooled sampling results appear in the main text and Sec. S3.

L	FidelityFormer raw	product approximation
24	0.082	0.061
32	0.130	0.080
48	0.209	0.114

Length-dependent error trends. The table reports length dependence before pooled affine OOD recalibration. Both predictors degrade as L grows, and the raw learned predictor has the larger bias at $L = 48$. The pooled affine correction reduces the overall length-OOD MAE to 0.077. The per-length values show that much of the raw bias comes from the longest sequences. A separate affine fit for each length reduces the $L = 48$ error further and requires labelled examples for every deployment length. The reported configuration uses one pooled fit and one shared label budget.

Per-seed (a, b) coefficients. Table S8 reports the affine coefficients for each `family_ood` seed. The slope and intercept have standard deviations 0.091 and 0.081. The post-calibration MAE has standard deviation 0.0013. Different coefficient pairs give similar predictive error.

Table S8: Affine recalibration coefficients (a, b) on `family_ood`, per seed. Linear fit $F_{\text{cal}} = a \cdot F_{\text{raw}} + b$ on a labelled, held-out 10% OOD calibration set.

seed	a	b	raw MAE	corrected MAE
0	2.845	−1.916	0.387	0.0952
1	2.822	−1.896	0.386	0.0949
2	2.918	−1.980	0.387	0.0937
3	3.026	−2.080	0.388	0.0954
4	3.003	−2.052	0.387	0.0923
mean	2.923	−1.985	0.387	0.0943
std	0.091	0.081	0.0008	0.0013

S6 Representation ambiguity and measurement-conditioned fusion

S6.1 Proof of the fibre-wise lower bound

Fix an observed representation x and choose $z, z' \in \mathcal{A}_\phi(x)$. Every estimator using only x returns the same value $h(x)$ on both instances. The triangle inequality gives

$$|F(z) - F(z')| \leq |F(z) - h(x)| + |h(x) - F(z')|.$$

At least one term on the right is at least $|F(z) - F(z')|/2$. Taking the supremum over pairs proves Eq. (8) of the main text. For absolute loss, the subgradient of $\mathbb{E}[|F - a| \mid x]$ changes sign at a conditional median. This establishes Eq. (9). The minimax quantity applies to the worst case within each fibre. The Bayes quantity averages over fibres under a stated distribution.

S6.2 Counterfactual protocol and random-stream control

We sample 2048 base collision sequences with lengths in $\{8, 16, 24\}$. For each base, we retain its couplings, frequencies, durations, bath initial state, and reset-bath marginal Choi sequence, then replay joint propagation at 15 equally spaced values in $\eta \in [0.85, 0.99]$. This gives 30,720 physical labels grouped into 2048 exactly identical observable inputs. The maximum tensor difference within every group is zero in float32 storage. Diameter is the within-group maximum minus minimum. The empirical Bayes estimate is the mean absolute deviation from the within-group median under the uniform 15-point grid.

To exclude pseudorandom coupling, we also generate a fresh 4096-instance evaluation set using seed 20260807 for collision parameters and a separate seed 920260807 for η. Its label mean and standard deviation are 0.4472 and 0.1979. On this split, 64-label affine BiDir calibration gives MAE 0.0894, summary ridge 0.0977, exact-marginal affine 0.1101, and an oracle ridge that additionally observes η 0.0800. These values reproduce the legacy split ordering without sharing a random stream.

S6.3 Full finite-label shot sweep

Every finite-label run uses 64 calibration instances. Independent 64-shot stratified-DFE estimates provide their target labels. For each budget B, a separate B-shot outcome on every calibration instance fits w_B in Eq. (11). A third random stream generates the test measurements. Target labels and pilot features come from different outcomes. All methods use the same test indices and test DFE outcomes. Neural entries aggregate five checkpoints and five measurement repeats. Other entries use five measurement repeats.

Table S9: Full independent-random-stream shot sweep. Values are MAE mean $\pm$ standard deviation. Offline shots for either hybrid are $4096 + 64B$.

B	stratified DFE	BiDir hybrid	summary-ridge hybrid
4	0.2755 ± 0.0029	0.0849 ± 0.0043	0.0946 ± 0.0051
8	0.1856 ± 0.0021	0.0810 ± 0.0025	0.0868 ± 0.0029
16	0.1299 ± 0.0015	0.0758 ± 0.0039	0.0805 ± 0.0050
32	0.0909 ± 0.0012	0.0657 ± 0.0019	0.0683 ± 0.0027
48	0.0743 ± 0.0013	0.0590 ± 0.0014	0.0605 ± 0.0016
64	0.0641 ± 0.0005	0.0539 ± 0.0012	0.0557 ± 0.0024
96	0.0525 ± 0.0004	0.0473 ± 0.0011	0.0485 ± 0.0024
128	0.0455 ± 0.0004	0.0424 ± 0.0016	0.0438 ± 0.0028

The mean neural fusion weight rises from 0.071 at four shots to 0.783 at 128 shots. At low budgets, the model supplies most of the estimate because the pilot has high variance. As the pilot becomes more precise, DFE receives more weight. Exact-label and finite-label calibration give similar curves. At high budgets, all priors converge toward DFE. The largest difference between neural and ridge priors occurs at low and intermediate budgets.